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Enhanced automated food making apparatus

Abstract

A method for operating an automated food making apparatus having a motor, actuator arm, and an apparatus. The apparatus may be a paddle with flexible fins. The method rotates the paddle with a pin-shaft mechanism to dispense an ingredient placed in a canister, controls the motor automatically based on weight sensor readings, and locates a position of the actuator arm with position sensors. The same motor dispenses ingredients from a plurality of canisters. The method may have a plurality of paddle rotation and weight measurement steps until a target weight is reached. The plurality of paddle rotation steps may be unidirectional or bidirectional paddle rotation. The paddle may be rotated according to one or more paddle rotation algorithms, an error recovery algorithm, or different algorithms based on the amounts of ingredients remaining in the canister. The paddle may be rocked until the target weight is achieved.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. Non-provisional patent application Ser. No. 16/933,130, filed Jul. 20, 2020, which will issue as U.S. Pat. No. 11,284,748, on Mar. 5, 2024, and is a continuation of U.S. Non-provisional patent

application Ser. No. 15/945,483, filed Apr. 4, 2018, which is a continuation-in-part of and claims the benefit of U.S. Provisional Patent Application No. 62/481,217, filed Apr. 4, 2017. U.S. Non-provisional patent application Ser. No. 15/945,483 is also a continuation-in-part of U.S. Non-provisional patent application Ser. No. 15/449,548, filed on Mar. 3, 2017, which issued as U.S. Pat. No. 10,813,503 on Oct. 27, 2020, and claims the benefit of U.S. Provisional Patent Application No. 62/304,277, filed Mar. 6, 2016. U.S. Non-provisional patent application Ser. No. 15/945,483 is also a continuation-in-part of U.S. Non-provisional patent application Ser. No. 14/847,959, filed on Sep. 8, 2015, which claims the benefit of U.S. Provisional Patent Application No. 62/047,785, filed Sep. 9, 2014, U.S. Provisional Patent Application No. 62/056,368, filed Sep. 26, 2014, U.S. Provisional Patent Application No. 62/094,595, filed Dec. 19, 2014, U.S. Provisional Patent Application No. 62/150,303, filed Apr. 21, 2015, U.S. Provisional Patent Application No. 62/185,524, filed Jun. 26, 2015 and U.S. Provisional Patent Application No. 62/201,105, filed Aug. 4, 2015. The contents of the foregoing applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) This application relates to the general field of electronically-aided apparatuses, systems, methods and techniques to conduct the food making process in a home or a business.

2. Discussion of Background Art

(2) Over the years, a number of innovations have come in to help with the cooking process. Food processors are now available to chop vegetables and meat. Induction cooktops allow a faster cooking process. Microwave ovens allow efficient reheating. However, despite these innovations, many of us spend an hour a day, or sometimes even more, cooking food for ourselves and our families. Cooking also requires a significant learning curve before one can do it in a tasty manner. As well, commercial food enterprises, such as restaurants, must currently allocate a significant amount of their costs on the human cooking endeavor. Ways to reduce the “human time” needed for cooking as well as the learning curve associated with cooking could be quite useful. As well, direct and indirect economic benefits may accrue for a business by transferring some of the human time costs to a machine, apparatus, robot, and such.

(3) U.S. Patent Application Publication No. 2013/0112683 from Hegedis, Davenport and Hoare apparently describes a cooking apparatus where a heating element works with a user interface and temperature sensors and provides prompts to the user during cooking. However, this requires user input to provide all the ingredients needed for cooking and requires the user to stand near the cooktop for large periods of time to respond to the prompts provided by the cooking apparatus. There is no mixing function available automatically, so the user needs to stand near the cooktop for large periods of time as well.

(4) U.S. Patent Application Publication No. 2011/0108546 from Cho and Chen apparently describes an intelligent heating mechanism which adaptively provides power to an induction cooktop based on temperature sensor data as well as a user-defined temperature profile. However, this requires the user to provide all the ingredients needed for cooking manually and requires the user to stand near the cooktop to mix the food items periodically.

(5) Foodini, a prototype and soon-to-be-released product from Natural Machines, apparently 3D prints food items by heating food pastes and dispensing them on to a stage. However, this requires food to be in paste form before being dispensed, which could be cumbersome and costly.

(6) Everycook, a prototype made in Europe, apparently promises to cut and mix food items and cook them with a recipe. However, the user still needs to be present near the Everycook cooking apparatus and dump additional food items every so often.

(7) Sereneti Kitchen, a prototype in the US, apparently wants to automate the cooking process but does not conduct any chopping of the ingredients and utilizes pre-chopped food instead. Neither does it put measured quantities of ingredients into the cooking vessel.

(8) What is needed are apparatuses and methods that allow food preparation with minimal human intervention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various embodiments of the invention will be understood and appreciated more fully from the following detailed description, taken in conjunction with the drawings in which:
- (2) FIG. 1 depicts an embodiment of this present invention, which may include a carousel atop the cooking pot;
- (3) FIG. 2 depicts the carousel mechanism shown in FIG. 1;
- (4) FIG. 3 illustrates an embodiment of this invention, wherein two carousels, one for housing ingredients and one for chopping ingredients, are placed atop a cooking pot;
- (5) FIG. 4 illustrates an embodiment of this invention wherein a container with a rotating dispenser knob is used in combination with the carousel mechanism of FIG. 1;
- (6) FIG. 5 illustrates an embodiment of this invention, namely an actuation mechanism for ingredient dispenser containers;
- (7) FIG. 6A illustrates an embodiment of this invention, which is an apparatus for chopping ingredients;
- (8) FIG. 6B illustrates an embodiment of this invention, which is an apparatus for dicing ingredients;
- (9) FIG. 7 illustrates an embodiment of this invention, which uses a series of links to move a stirrer to various positions;
- (10) FIG. 8 illustrates an embodiment of this invention, which may dispense solid ingredients;
- (11) FIG. 9 illustrates an embodiment of this invention, wherein food is prevented from sticking on to the sides of the ingredient container by reducing the surface area of contact between the ingredient container and the food;
- (12) FIG. 10A and FIG. 10B illustrate embodiments of this invention, wherein the mechanism for dispensing and sensing is described;
- (13) FIG. 11 illustrates an embodiment of this invention, wherein the mechanism for dispensing food is described;
- (14) FIG. 12A and FIG. 12B illustrate embodiments of this invention that dispense liquids;
- (15) FIG. 13 illustrates an embodiment of this invention that shows a mass sensor system;
- (16) FIG. 14 illustrates an embodiment of this invention that shows a system capable processing various types of food;
- (17) FIG. 15 illustrates an embodiment of this invention that shows the system for placing the salad bowl or pizza base or cooking pot and heater or tortilla (for making burritos) and in general for placing the base that is processed further;
- (18) FIG. 16 illustrates an embodiment of this invention that illustrates a modular ingredient container and shows how it may be attached to a carousel;
- (19) FIG. 17 illustrates an embodiment of this invention, that shows how modular ingredient containers may be attached to each other;
- (20) FIG. 18 illustrates an embodiment of this invention, a paddle for dispensing ingredients;
- (21) FIG. 19 illustrates an embodiment of this invention, a bearing for ingredient containers;
- (22) FIGS. 20A-20C illustrate an embodiment of this invention, which show how magnets and Hall sensors may be structured for dispensing material from ingredient containers;
- (23) FIG. 21 illustrates an issue with a proposed dispensing system, wherein vertical knobs can collide with an actuator used for dispensing;
- (24) FIG. 22 illustrates an embodiment of this invention, that shows how a knob may be

straightened with a “knob straightener mechanism”;

(25) FIGS. **23A-23B** illustrate an embodiment of this invention, that shows how a robot may be controlled using a touchscreen user interface;

(26) FIGS. **24A-24B** illustrate an embodiment of this invention, which show how thermal insulation is provided between the chamber where ingredients are kept and other portions of the apparatus;

(27) FIG. **25** illustrates an embodiment of this invention, which shows how a container may be used for providing thermal insulation by closing the hole where ingredients drop;

(28) FIGS. **26A-26C** illustrate an embodiment of this invention, which show a mechanism for opening and closing a hole where ingredients drop;

(29) FIG. **27** illustrates an embodiment of this invention, which shows a method to dispense ingredients;

(30) FIG. **28** illustrates an embodiment of this invention, which shows a method to dispense liquid ingredients;

(31) FIG. **29** illustrates an embodiment of this invention, which shows an Ingredient Cut Based Dispensing algorithm;

(32) FIG. **30** illustrates an embodiment of this invention, which shows a Paddle Based Dispensing algorithm;

(33) FIG. **31** illustrates an embodiment of this invention, which shows a Threshold Based Speed algorithm;

(34) FIG. **32** illustrates an embodiment of this invention, which shows a Threshold Based Weight Measurement Frequency algorithm;

(35) FIG. **33** illustrates an embodiment of this invention, which shows an Ingredient Level Based Dispensing algorithm;

(36) FIG. **34** illustrates an embodiment of this invention, which shows a Liquid Pullback algorithm;

(37) FIG. **35** illustrates an embodiment of this invention, which shows a Dispenser Collision Recovery algorithm;

(38) FIG. **36** illustrates an embodiment of this invention, which shows an Ingredient Jamming Recovery Reverse Direction algorithm;

(39) FIG. **37** illustrates an embodiment of this invention, which shows an Ingredient Jamming Recovery Carousel Shake algorithm;

(40) FIG. **38** illustrates an embodiment of this invention, which shows a Fallback Container algorithm;

(41) FIG. **39** illustrates an embodiment of this invention, which shows a Rocking Motion Dispense algorithm;

(42) FIG. **40** illustrates an embodiment of this invention, which shows a Bidirectional Motion algorithm;

(43) FIG. **41** illustrates an embodiment of this invention, which shows a Switch Directions Between Salads algorithm;

(44) FIG. **42** illustrates an embodiment of this invention, which shows a Quantized weight dispense algorithm;

(45) FIG. **43** illustrates an embodiment of this invention, which shows a Multiple Ingredient Dispensing algorithm;

(46) FIG. **44** illustrates an embodiment of this invention, which shows a Predictive Zero Mechanism Undershoot algorithm;

(47) FIG. **45** illustrates an embodiment of this invention, which shows a Predictive Dispensing Undershoot algorithm.

(48) FIG. **46A-D** illustrate an embodiment of this invention, which show a liquid dispensing mechanism

(49) FIG. **47A-C** illustrate an embodiment of this invention, which show tabbed paddles

- (50) FIG. **48A-C** illustrate an embodiment of this invention, which show shufflers for dispensing ingredients that don't operate perfectly under gravity fed mechanisms
- (51) FIG. **49A-D** illustrate an embodiment of this invention, which show an apparatus which snaps a pin shaft mechanism and a paddle onto a canister
- (52) FIG. **50** illustrates an embodiment of this invention, which describe a rocking motion dispense algorithm with weight feedback.

DETAILED DESCRIPTION

(53) Embodiments of the present invention are now described with reference to at least the above figures. Persons of ordinary skill in the art will appreciate that the description and figures illustrate rather than limit the invention and that in general the figures are not drawn to scale for clarity of presentation. Such skilled persons will also realize that many more embodiments are possible by applying the inventive principles contained herein and that such embodiments fall within the scope of the invention which is not to be limited except by any appended claims.

(54) FIG. **1** describes an embodiment of this present invention, which may be a robotic cooking apparatus or food preparation machine/apparatus. A robotic cooking apparatus may include outer container **100**, inner container **102**, carousel **104**, shaft **106**, pan **108**, stirrer **110**, robot arm **112**, X rail **114**, Y rail **116**, motor **118**, plate **120**, and heater **122**. Food may be stored in ingredient dispenser containers such as outer container **100** and inner container **102**. The terms tube and canister may also be used to refer to containers at various sections of this patent application. Ingredient dispenser containers outer container **100** and inner container **102** may be mounted on to a carousel **104** which may be attached to a rotating shaft **106**. The shaft **106** may be rotated with the help of a motor. Several mechanisms may be used for rotating containers placed in a circular configuration, which may be placed on a circular board/platform. In FIG. **1**, two circular rows of ingredient dispensers are depicted, with outer container **100** on the outer circular row and inner container **102** on the inner circular row. A number of circular rows may be designed and utilized, and may range from at least 1 to 10. The carousel **104** may be placed atop the pan **108** where cooking may happen. Pan **108** may be called pot, cooking pot, cooking pan, or cooking vessel herein. The carousel **104** may include openings (not shown), including substantially circular and other shapes, for dispensing food from ingredient containers outer container **100** and inner container **102**, and other containers. These circular openings may be structured such that when food falls through these circular openings, they fall into the pan **108**. A heater, such as, for example, an induction heater **122**, may be used to cook a dish. It may include a stirrer **110** which may be moved in X and Y dimensions (with respect to pan **108**) using robot mechanisms which may include round shafts or rails such as X rail **114** and Y rail **116**. Stirrer **110** may also be designed to move in the Z dimension and various angles/combinations of X, Y and Z. A motor **118** may be used for rotating the stirrer **110**. Several variations of these embodiments are possible. For example, the stirrer **110** may be attached to a polar robot mechanism. Polar mechanisms may provide improved resistance to cooking grease related reliability issues since they may be easier to seal. Cooking pan **108** and heater **122** may be moved via moving plate **120** up and down using a robot arm **112**. Robot arms shown in FIG. **1** may be built using a number of different mechanisms, such as, for example, chains, belts, lead screws, ball screws and many other materials. A refrigeration system, a Peltier cooling system or other cooling apparatus may be utilized to cool the region above the carousel **104**, and efficiency improved by placing components above the carousel **104** in a thermally isolated environment. The opening on the carousel which may allow food to be dispensed into the pan **108** may be open and closed using a robot arm or other actuation mechanism. The plate **120** may include a mass sensor which measures the weight of food in the pan. This may provide information about the status of a certain dispensing step i.e. how much food has been dispensed from ingredient dispensers such as outer container **100** and inner container **102** into the pan **108**. The mass sensor may optionally also provide information about status of the cooking process—by measuring how much weight reduction happens during the cooking process. It will be clear to one skilled in the art

that several variations of these embodiments may be possible. For example, an induction heater need not be present, and one may dispense ingredients using the robotic cooking apparatus for making salads and other types of food. Sensors (not shown) may be present for estimating if ingredients in containers such as inner container **102** may be spoilt. Carousel **104** may include more than two rows of containers or just one row of containers. The temperature of the environment in which the carousel with containers is placed can be modulated, for example, using a refrigeration system or a heating system.

(55) FIG. **2** illustrates a close up view of a design of the carousel described in FIG. **1**. Outer containers **200** and inner containers **202** may be placed on the carousel **204** which may contain a shaft **206**. Placement of outer containers **200** and inner containers **202** on carousel **204** may be designed such that their bottom opening may be positioned substantially directly over opening(s) (not shown) in the thermally isolated carousel environment of FIG. **1**. A chute configuration (not shown) may alternatively be employed wherein the containers are not substantially directly over the opening(s). Gravity feed as well as motorized movement of the food ingredients from the containers thru the opening(s) to the pan (or other receptacle) may be utilized.

(56) FIG. **3** illustrates an embodiment of this present invention wherein two carousels, upper carousel **300** and lower carousel **302**, may be placed above a cooking pan (not shown). The upper carousel **300** may be connected to containers having ingredients, such as outer ingredient container **304** and inner ingredient container **306**. The lower carousel **302** may be connected to choppers such as chopper **308**. Some choppers may contain blades to slice ingredients, some choppers may contain blades to dice ingredients, some choppers may contain blades to shred ingredients and some choppers may have other functions. The robotic cooking apparatus can control which ingredient container is placed above which chopper by rotating individual carousels upper carousel **300** and lower carousel **302**, such that a certain ingredient, or combination of ingredients, may be chopped. Several mechanisms may exist to rotate carousels upper carousel **300** and lower carousel **302**. For example, belts such as upper belt **312** and lower belt **318** may be used, in combination with pulleys upper carousel pulley **310**, upper motor pulley **314** and lower motor pulley **316**. Direct drive and other gearing mechanisms may also be utilized to rotate upper carousel **300** and lower carousel **302**.

(57) FIG. **4** illustrates an embodiment of this invention, wherein the container shown in FIG. **4** may be used in conjunction with the carousel mechanism of FIG. **1** to dispense controlled amounts of ingredients. View **400** shows a side view of the container that may be used in the carousel **104** while second view **402** shows an exploded view of the container that may be used in the carousel **104**. The container may include an object such as cylinder **404** for housing the ingredients. Cylinder **404** may have a square or rectangular cross sectional shape, the diameter may increase or decrease in the vertical direction, and the material composition and surface friction coefficient/roughness chosen, depending on design and engineering considerations, for example of the food ingredient type, moisture content, container cleaning/sterilization constraints and so on. Shapes such as container side **406** may be added to make insertion into a carousel mechanism easier by inserting the shapes into slots on a carousel. A shape such as handle **408** may be used to dispense controlled amounts of ingredients. The exploded second view **402** shows more details of the ingredient dispensing mechanism. When the knob **410** is rotated, the shaft **414** may rotate the paddle **412**. The rotational motion may allow dispensing of controlled amounts of ingredients. Paddle **412** may be partially constructed of a flexible material, for example, such as silicone. Mass sensors (not shown) may be used in conjunction with this mechanism to determine the amount of ingredient dispensed. In addition, monitoring the rotational angle (theta) traversed by the knob **410** may provide an estimate/measure of the ingredient amount dispensed.

(58) FIG. **5** describes an embodiment of this invention, which illustrates an apparatus for actuating the knob **410** of container cylinder **404** in FIG. **4** herein. Knob **402** (or some other projection) of the dispenser container may be present, and may be indicated as projection **502**. To rotate the

projection **502**, a gripper mechanism may be used. Two arms of the gripper upper arm **504** and lower arm **506** may be used to grip and then securely hold the projection **502**. Following this, the motor **510** may be used to rotate the gripper by rotating gripper body **508**. In case some food items get stuck in container cylinder **404**, gripper body **508** may be rotated in the opposite direction. Motor **510** and hence gripper body **508** (and ultimately the paddle **412**) may also be run thru an acceleration/deceleration forward/reverse algorithm (for example, creating a vibration) to clear stuck food items. Several other mechanisms are possible to hold and rotate the projection **502**, for example, utilizing a robotic arm, or single/quad gripper arms.

(59) FIG. **6A** illustrates an embodiment of this invention, which is an apparatus for chopping ingredients in the carousel mechanism that may be depicted with FIG. **1**. Exemplary ingredient containers **600** may be placed in carousel **602**. Chopping sliders **604** may be placed into sockets **606** at the base of ingredient containers so that they can slide back and forth in the sockets **606**. Chopping blades **608** may chop up ingredients in containers when the chopping sliders **604** are moved in a certain direction. Chopping sliders **604** may be pushed and pulled using an actuator mechanism (not shown in the figure).

(60) FIG. **6B** describes an embodiment of this invention, which is an apparatus for dicing ingredients in the carousel mechanism that may be depicted with FIG. **1**. Exemplary ingredient containers **620** may be placed in carousel **622**. Chopping sliders **628** may be placed into sockets **630** at the base of ingredient containers so that they can slide back and forth in the sockets **630**. Dicing grids, such as for example, **624** may be placed at the base of ingredient dispensers. Ingredients may be pushed down the ingredient containers using a plunger mechanism, such as for example, the plunger described. The action of ingredients being pushed down the ingredient dispenser into the dicing grid, in combination with the motion of chopping sliders **628**, together may cause ingredients to be diced and dispensed. Chopping sliders **628** may also include chopping blades **626** to provide a dual use function.

(61) FIG. **7** illustrates an embodiment of this invention which allows motion of components in a plane based on motion of multiple links first link **706** and second link **708**. Motors first link motor **700** and second link motor **702** could be used to rotate links first link **706** and second link **708** and thus move a stirrer **710** to various points in a cooking vessel **714**. Stirrer motor **704** may be utilized to provide other motions of stirrer **710**, for example, rotation clockwise and counterclockwise, specific stirrer blade orientations in combination with link movement and orientation (for example, to provide a scrapping action on the surface of cooking vessel **714**), and so on. The cooking vessel **714** may be located atop a heater **716**. With this type of robotic system for handling the stirrer **710**, wires and motors may be enclosed and thereby protected from environmental factors, such as, for example, dirt and grease. This type of link based system can be used for moving or providing motion to objects and mechanisms other than stirrers, such as, for example, spice dispensers, liquid dispensers and other objects. Several variations of this link based system may be possible. For example, one could have more links than two, motors may be placed at alternative positions, Z motion and combinations of X, Y, and Z motions, and many other options may be possible.

(62) FIG. **8** illustrates an embodiment of this invention, a solid dispensing apparatus. A paddle **806** (similar to paddle **412** of FIG. **4** herein) may be present within a food containing tube **802** (which is similar to ingredient containers of at least FIGS. **1-4**, **6A** and **6B** herein). The tube **802** may be attached to a carousel using collars **804**. The knob **808** (similar to knob **410** of FIG. **4** herein) may be rotated using the help of a motor to rotate the paddle **806** and in combination with gravity, dispense food. The term pin may also be used to describe the knob at various sections of this patent application. To reduce sticking of food in the food containing tube **802**, the knob **808** may be rotated in more than one direction during the dispensing process, as described previously in at least FIG. **4** & related specification sections herein. At various points of this patent application, the terms tubes and canisters may be used interchangeably.

(63) FIG. **9** illustrates an embodiment of this invention, which may help reduce sticking of food on

the sides of container **802** depicted in FIG. **8**. This may be done by having non circular sidewalls **912** on the inside of the container so that surface area of contact between the food item and the inside wall is reduced. The outside wall **910** may be circular. Several variations of these embodiments may be possible. For example, one could have non-circular inner and outer walls and one may use wave-like patterns or other patterns on the inner walls to reduce sticking. The pattern could be tuned or 'matched' to the type and shape of the food ingredient. For example, a vertical wave pattern could be the half or quarter period of the average size ('wave') of the food item.

(64) FIG. **10A** and FIG. **10B** illustrate an embodiment of this invention, a mechanism to rotate the knob **808** shown in FIG. **8**. In FIG. **10A**, the motor **1002** may be used to rotate a shaft **1008** which may in turn rotate the dispensing mechanism **1006**. A magnet may be used as part of the dispensing mechanism **1006**. A hall sensor **1010** shown in FIG. **10B** may be used to determine the rest position of the knob **808** after the dispense operation is complete.

(65) FIG. **11** illustrates an embodiment of this invention, a mechanism to dispense food, which may include ingredient container **1100**, ingredient container knob **1102**, dispensing knob **1104** and motor **1106**. A motor **1106** may be used to rotate the dispensing knob **1104**. When dispensing knob **1104** rotates, the ingredient container's knob **1102** may also rotate. This, in turn, may dispense food ingredients from ingredient container **1100**. The term "pin" may be used instead of the term "knob" at various sections of this document.

(66) FIG. **12A** and FIG. **12B** illustrate embodiments of this invention, a liquid dispensing system, which may include pin **1202**, ingredient container **1204**, spacer **1206**, cam mechanism **1208**, shaft **1210**, ingredient container knob **1212**, pin **1214**, head **1216** and spout **1218**. When the ingredient container knob **1212** may be rotated, the cam mechanism **1208** may be pushed up on the spacer **1206**. When the cam mechanism **1208** is pushed up, the spout **1218** may dispense the ingredient from the container **1204** using a pump mechanism. A one way valve may be added to the end of the spout **1218** to reduce dripping of the liquid when dispensing action is not required.

(67) FIG. **13** illustrates an embodiment of this invention, a mass sensor scheme, which may include load cell **1302**, mass measuring system **1304** and bowl **1306**. A load cell **1302** may be used and attached to a mass measuring system **1304**. When food falls into the mass measuring system **1304** through the top opening into the salad bowl **1306**, the weight may be measured. Based on whether the desired weight of the ingredient has been dispensed, the motor for dispensing the ingredient may be turned to OFF position. The mass sensor system shown in FIG. **13** is isolated from the food zone where the salad bowl or cooking container or induction heater may be placed. According to an embodiment of this invention, the bowl **1306** may be placed such that it is isolated from wires associated with the load cell **1302**.

(68) FIG. **14** is an illustration of an embodiment of this invention which illustrates a food system **1499**, part of a robotic cooking apparatus, capable of helping make pizza, cook food, make burritos, make salads and make several other types of food. Food system **1499** may include plate **1402**, second link motor **1404**, first link motor **1406**, compartment **1408**, ingredient containers **1410**, carousel **1412** and dispenser motor **1414**. Ingredients may be placed in ingredient containers **1410** (one shown for clarity) and may be dispensed using motion of the carousel **1412** and dispensing mechanisms which use dispenser motors such as, for example, dispenser motor **1414**. The dispensing mechanisms may be shared among multiple containers to lower cost and weight of the food making machine.

(69) In case of making pizza, the pizza base may be placed on the plate **1402**. The plate **1402** may be moved using a multi-link mechanism which in turn may move based on motion of motors second link motor **1404**, first link motor **1406** and additional motors placed in compartment **1408**. Ingredients may be dropped on the pizza base using technologies described in FIG. **1**-FIG. **13** herein. The pizza base may be moved using motion of the plate **1402** to distribute the ingredients over the pizza area.

(70) In case of making burritos, the tortilla may be placed on the plate **1402** and ingredients may be

dispensed atop it.

(71) In case of making salads, the salad bowl may be placed on the plate **1402** and ingredients may be dispensed atop it.

(72) In case of making one pot meals, such as, for example, stews and many Indian and Chinese and Thai entrees, an induction heater and the pot may be placed atop the plate **1402** and ingredients may be dispensed into the pot. An additional robot arm may be used to stir the food. The robot arm may be designed as a Cartesian robot system with a stirrer at the end or using a technique similar to the one described in FIG. 7 herein or using some other technique.

(73) FIG. 15 is an illustration of an embodiment of this invention, which illustrates a closer view of the mechanism for moving the plate **1402** of FIG. 14. The plate **1502** may be moved using motion of links third link **1506**, second link **1510** and first link **1512**. The motors third link motor **1504** and second link motor **1508** may rotate to move links third link **1506** and second link **1510** and thereby move the plate **1502** in the horizontal plane. The first link **1512** may move up and down via motors placed within compartment **1514**. Several other mechanisms may provide movement to the plate **1502** in the X, Y, Z plane and dispense ingredients onto it. For example, placing plate **1502** on a 3D motion table.

(74) FIG. 16 is an illustration of an embodiment of this invention that describes a modular ingredient container and shows how it may be attached to a carousel. The modular ingredient container **1642** (and blow-up **1640**) may consist of two or more portions (such as, for example, upper portion **1623** and lower portion **1624**) that may be attached to each other using a latch mechanism **1644**. Using modular ingredient containers is an innovation that provides several benefits: (1) If one wants to increase food capacity of the apparatus, one more modular ingredient container portions can be added to provide extra capacity (2) A big size ingredient container, when split into two smaller ingredient containers, is easier to fit into a dishwasher or sink for cleaning purposes. The modular ingredient containers may be attached to carousel **1625** using various mechanisms. These may include a pin-mechanism, wherein pins such as pin **1630** may be inserted into slots such as left slot **1619** and right slot **1626**. Modular ingredient containers may also be attached to carousel **1625** using clip mechanisms, wherein a clip **1628** may be used to attach to a portion of an ingredient container such as location **1620**. An example where a portion of an ingredient container is attached to a clip **1622**. Several alternative mechanisms may be possible to attach an ingredient container to the carousel. For example, magnets, for example a combination of permanent and electromagnets may be used. Pins, such as for example, cotter pins **1632** may be used to make sure a shaft used in the canister does not slip out.

(75) FIG. 17 is an illustration of an embodiment of this invention how different portions of an ingredient container may be attached to each other. Projections such as first projection **1712**, second projection **1713**, third projection **1710** and fourth projection **1714** may be added to ingredient container portions upper portion **1717** and lower portion **1716** which may need to be attached to each other. A joiner may be added that may consist of parts such as flap **1715**, elastic flap **1711** and stem **1720**. The elastic flap **1711** may allow for a good fit despite the manufactured tolerance of various parts. It may consist of a flexible material that can deform to allow a good fit. Examples of flexible materials may include silicone rubber, polyurethane and many other materials. The stem **1720**, flap **1715** and other parts of the joiner may consist of non-flexible materials so that multiple parts of an ingredient container are closed securely, without leakage of material. Examples of materials for this application may include polycarbonate, PVC and many other materials. The ingredient containers may be open or closed by moving joiners into open or closed positions. FIG. 17 includes locked position **1718** and unlocked position **1719** illustrations.

At various sections of this patent application, the term latch may be used in place of the term joiner.

(76) FIG. 18 is an illustration of an embodiment of this invention how paddles may be designed for use in the ingredient containers. The paddles may, for example, be constructed of similar or multiple different materials for core **1834** and external portions first extension **1830** and second

extension **1831**. According to one embodiment of this invention, the core **1834** may primarily include a non-flexible plastic, such as, for example, polycarbonate, PVC or other suitable non-flexible plastic. The external portions first extension **1830** and second extension **1831** may have flexible materials, such as, for example, silicone rubber, polyurethane or some such material. According to one embodiment of this invention, the external portion first extension **1830** may be thicker than the external portion second extension **1831**. This could provide the most effective combination of stiffness and flexibility for dispensing specific ingredients. Alternatively, one could have just one thickness for the whole external portion. It will be clear to one skilled in the art that several different thicknesses for the non-flexible plastic may be possible at different external portions of the paddle, to provide various mechanical properties needed for dispensing ingredients. According to an embodiment of this invention, the external portions first extension **1830** and second extension **1831** may be overmolded atop the core **1834**. Holes **1832** may be inserted into the core **1834** to allow more convenient overmolding.

(77) FIG. **19** is an illustration of an embodiment of this invention how bearings may be used to provide long-term reliability to containers. When shafts **1933** are inserted into containers **1936** and rotated over long-periods of time to dispense ingredients, plastic used in containers **1936** may degrade and/or wear away. By inserting bearings outer bearing **1940** and inner bearing **1938** into the ingredient container **1936**, the reliability challenges may be reduced. Various types of bearings and materials for bearings may be possible, and could reduce friction, degradation or wear.

(78) FIGS. **20A-20C** illustrate an embodiment of this invention, wherein multiple hall sensors and magnets may be placed within a dispenser motor assembly to more accurately dispense ingredients. FIG. **20A** indicates a dispensing actuator arm **2004**, a motor shaft **2006** that rotates actuator arm **2004**, a plate **2008** and a motor cover **2002**. Two Hall Sensors sensor one **2010** and sensor two **2012** may be used to detect the location of the actuator arm **2014** based on position of magnets top magnet **2016** and bottom magnet **2018**. When a magnet is directly above a sensor during rotational motion of actuator arm **2014**, the sensor may indicate it and give feedback to the control PCB on location of the actuator arm. Various types of sensors may be possible, not just Hall sensors. Magnets may be of various shapes, sizes and types. More than two Hall sensors may be used. A single Hall sensor architecture may be used as well. Alternatively, an encoder may be used in the motor to indicate its position.

(79) FIG. **21** illustrates an issue that arises when using a pin-dispenser rod actuator system **2106**. The pin **2102** and actuator arm **2104** may be aligned in the same direction and could crash during motion of a carousel. This needs to be avoided for proper system operation. FIG. **22** illustrates an embodiment of this invention, a system for aligning pin **2204** so that it does not collide with the actuator arm shown in FIG. **21**. A pin straightener **2202** may be placed in the apparatus. When the carousel rotates, pins **2204** may be automatically aligned into a horizontal direction due to engagement with the pin straightener **2202**.

(80) FIGS. **23A-23B** illustrate an embodiment of this invention, wherein a touchscreen may be used to control the operation of a food preparation/robotic cooking apparatus having one of more of the features indicated in FIGS. **1-22** and FIGS. **24-28**. The touchscreen **2308** may be placed within door **2306**, as indicated in FIG. **23B**. FIG. **23A** indicates the back side of the door **2304** and view **2302** indicates an exemplary carousel system with exemplary canisters loaded on it. Customers may use the touchscreen **2308** to indicate their food choices and the apparatus indicated in FIGS. **23A-B** may prepare the food.

(81) Food preparation apparatuses as shown in this patent application frequently need to be refrigerated to store food over prolonged periods of time without spoilage. FIGS. **24A-B** illustrates an embodiment of this invention, which is a system for thermally insulating the food storage chamber of the apparatus. The system may consist of an insulation canister **2404** meant for insulation purposes. One position of the insulation canister **2404** may be indicated in FIG. **24A**, where the insulation layer **2406** does not contact the food opening **2402** i.e. the food opening is

unsealed. Another position of the insulation canister **2404** may be indicated in FIG. **24B** wherein the insulation layer **2406** may contact the food opening **2402**, seal it and prevent significant heat from entering the chamber. The insulation layer **2406** may include a good insulator, such as, for example, silicone or some other insulation material. The insulation layer **2406** may also include a material which has some flexibility so that it gives a tight fit to the food opening **2402**. When the apparatus is not being used for making food, a carousel may move the canister meant for insulation (insulation canister **2404**) directly above the food opening **2402** and keep the food storage chamber insulated. It will be clear to one skilled in the art that several variations of this embodiment may be possible. For example, shape of canisters, insulation layers and food openings may be different than illustrated. Insulation canister may also contain some insulative material in addition to insulation layer **2406**.

(82) FIG. **25** illustrates different portions of the insulation canister described in FIGS. **24A-24B**. The canister may be composed of two portions, upper portion **2502** and lower portion **2504**, for example. The insulation layer **2506** may be connected to a mechanism within the insulation canister using pieces **2508**. FIGS. **26A-26C** illustrates a simplified view of the internal mechanism within an insulation canister. It will be clear to one skilled in the art that the mechanism shown in FIG. **26A-26C** is exemplary, and several variations may exist. An insulation layer **2606** may be connected to a platform **2604** that moves within a canister. A pin **2610** may be rotated with a dispensing actuator similar to those described earlier in this patent application. The pin may actuate a mechanism consisting of a cam **2614** using a shaft. FIG. **26B** may illustrate one position of the mechanism, where the portion **2616** of cam **2614** may come in contact with the wall **2618**. A wheel **2612** may allow smooth motion of the cam **2614**. The platform **2604** is not shown in FIG. **26B-26C** to better illustrate the working of the mechanism. FIG. **26C** may illustrate another position of the mechanism, where the cam **2620** may be in another stable position. One of the key factors of the invention shown in FIG. **26A-26C** is the fact that the cam **2614** may be in two stable positions. This provides a stable open and closed position of insulation layer **2606**, 'closed' with respect to food opening **2402** when actuated 'down', and 'open' with respect to food opening **2402** when the cam position pulls insulation layer **2606** 'up' so that the insulation canister **2404** can rotate freely on the carousel. Thus the insulation canister may be operated by the same motor/cam system as the normal food dispensing operation.

(83) FIG. **27** illustrates an embodiment of this invention, wherein ingredients sticking to the walls of the ingredient container/canister may be reduced by using fittings **2704** within the canister. These fittings may be actuated by motion of the paddles **2710**. The fittings **2704** may be attached to the top of canisters **2708** or the sides of canisters **2709**. They may have multiple pieces, for example, with one portion fitting bottom **2707** contacting the paddle and another portion fitting top **2704** contacting top of canisters **2708**. When the paddle rotates, it may move the fittings back and forth by contacting fitting bottom **2707** and causing motion within the canister which may allow ingredients stuck to the sides of canisters to come unstuck. Snapshot one **2700** illustrates fitting **2704** with no paddles **2710** contact, snapshot two **2701** illustrates fitting **2704** contacting paddles **2710** to one side, snapshot three **2702** illustrates fitting **2704** contacting paddles **2710** to the other side. Several variations of this embodiment may be possible. For example, the shape of the fittings may be different—it could be the shape of a curtain. The fittings may be attached to the sides of the canister instead of the center as shown in FIG. **27**. The fittings may include a hinge. Several other variations may be possible.

(84) FIG. **28** illustrates an embodiment of this invention, which shows an apparatus for dispensing liquids. The liquid to be dispensed may be stored in a bottle located within a canister **2806** and a flexible tube **2800** may lead from it. The flexible tube may be compressed by rollers such as **2802** and **2804** to control the dispensing of the liquid. A one way valve may be added to the end of the tube **2810** to reduce dripping of liquid in unwanted locations. The rollers **2802** and **2804** may move using rotation of shaft **2812**, which in turn may be rotated using a shared dispensing that may be

connected to a pin **2812** located on the canister **2806**.

(85) Additional Methods, Algorithms and Software.

(86) The apparatus of the automated food making machine, for example FIG. **14** herein, and sub-apparatuses, for example pin straightener **2202** of FIG. **22** herein, may be controlled by a computer system wherein various algorithms and software instantiated in the computer/microprocessor system may form methods of operation and control of the machine or sub-apparatuses. Following are inventive embodiments of the methods, algorithms and software. Of course, some of these functions may be controlled from a computer/microprocessor that is not within the food making machine, for example, a centralized control system at a corporation, home, or operated from/by the manufacturer.

(87) The algorithms and software programs may include at least the following commands and values:

(88) TABLE-US-00001

Min	Max	Name	Command	Description	value	value	Units	Container	C
1	24	to be selected	Speed S	rotational speed of	0	600	rpm	dispenser	Target W
un-adjusted weight	0	200 grams	Weight to be targeted by machine	Angle A	angle to move	0	2160		
before taking weight measurement	No. of Q	number of	1	50	samples	weight measurements			
Samples taken and averaged to reduce noise	Timeout T	number of times the	1	30	loops	Loops	loop		
runs before flagging an error	Undershoot U	percentage to	0	50	%	% change target weight by to			
account for bias	Undershoot V	quantity to change	0	20	grams	Quantity target weight by to account for bias			
Motion G	which motion	1	n	algorithm	Algorithm	algorithm to use	number		

(89) The min and max values may be adjusted based on engineering and design considerations. For example, the No. of weight Samples, Q, may have a max value greater than 50 if a faster reading scale is used for a particular overall machine model.

(90) For example, an algorithm and software program may have the following steps:

(91) TABLE-US-00002

Command & Step	Description	Value	Step
1	Set Container to #6	M6 C6;	
2	Set Target Weight to 50 g	W50	Step 3
3	Set Motion Algorithm	G1 A180 S500 (angle, speed)	
4	Set Undershoot % to 50%	U50	Step 5
5	Set Motion Algorithm	G2 A90 S100 (angle, speed)	
6	Set Undershoot % to 90%	U90	

(92) The above example may be written as three lines of code as follows:

(93) TABLE-US-00003 1 M6 C6; 2 W50 G1 A180 S500 U50; 3 G2 A90 S100 U90;

(94) An Ingredient Cut Based Dispensing algorithm and software program may be disposed in the food making machine apparatus, and may select and control different cuts (for example; julienne, dice, chopped, etc.) for various ingredients (for example; iceberg lettuce, romaine lettuce, carrots, beets, cheese, etc.) which may require different sub-algorithms to control the appropriate machine sub-units and/or components. As illustrated in FIG. **29**, an illustrative example of an Ingredient Cut Based Dispensing algorithm and software program is shown in an overview flowchart. For example, the Start **2900** may begin the algorithm and the first question asked may be is the ingredient to be diced [**2910**]. For example, the customer may order diced cucumber for the salad, so the machine may be directed to the cucumber container and use a dicing sub-algorithm 1 [**2930**] to actuate the dicing apparatus under the cucumber container. If the ingredient is to be sliced [**2912**], then dispensing algorithm 2 [**2932**] may be utilized to move and operate the slicing and dispensing mechanism for that ingredient (for example, to slice the cucumbers from the cucumber container). If the ingredient is to be shredded [**2914**], then dispensing algorithm 3 [**2934**] may be utilized to move and operate the shredding and dispensing mechanism for that ingredient. If the ingredient is to be chopped in some other form [**2916**], then dispensing algorithm 4 [**2936**] may be utilized to move and operate the shredding and dispensing mechanism for that ingredient. If the ingredient is to be treated in some other way (a No to [**2916**]), then dispensing algorithm 5 [**2938**] may be utilized to move and operate the appropriate mechanism for that ingredient. All of the dispensing algorithms, when the appropriate amount of the ingredient is dispensed, may conclude with End [**2999**].

(95) A Paddle Based Dispensing algorithm and software program may be disposed in the food making machine apparatus, and may select and control different paddles (for example; 2 fin, 4 fin, 6 fin, flexible, rigid, rig/flex, etc.) for various ingredients (for example; iceberg lettuce, spinach, carrots, nuts, raisins, seeds, croutons, etc.) which may require different algorithms to control the appropriate machine sub-units and/or components. As illustrated in FIG. 30, an illustrative example of a Paddle Based Dispensing algorithm and software program is shown in an overview flowchart. For example, the Start [3000] may begin the algorithm and the first question asked may be is the ingredient diced [3010]. For example, the customer may order diced cucumber for the salad, so the machine may be directed to the cucumber container and use a dicing sub-algorithm (see FIG. 29). Then dispensing Paddle type 1 [3030] may be actuated to precisely dispense the diced food. If the ingredient is sliced [3012], then dispensing Paddle type 2 [3032] may be precisely dispense the sliced food. If the ingredient is shredded [3014], then dispensing Paddle type 3 [3034] may be utilized to precisely dispense the shredded food for that ingredient. If the ingredient is to be chopped in some other form [3016], then dispensing Paddle type 4 [3036] may be utilized to precisely dispense the chopped food for that ingredient. If the ingredient is to be treated in some other way (a No to [3016]), then dispensing Paddle type 5 [3038] may be utilized to precisely dispense the food for that ingredient. All of the dispensing paddle type algorithms, when the appropriate amount of the processed ingredient is dispensed, may conclude with End [3099].

(96) A Threshold Based Speed algorithm and software program may be disposed in the food making machine apparatus, and may select and control different dispense rates/speed as a function of another input (for example; a faster food dispense speed until 80% of target weight, then a slower speed to finish, etc.). The input, for example, may be the weight of food dispensed, and the sample rate of that weight may be adjusted in a certain manner, for example, inversely proportional to the percentage of the target weight, and so on), and may be different (speed control and weight sampling rate) for various ingredients (for example; iceberg lettuce, spinach, carrots, nuts, raisins, seeds, croutons, etc.) which may require different algorithms to control the appropriate machine sub-units and/or components. As illustrated in FIG. 31, an illustrative example of a Threshold Based Speed algorithm and software program is shown in program form and an overview flowchart. For example, the Start [3100] may begin the algorithm and a default speed S1 of the dispensing motor rotation [3110] may be set to dispense the food, whilst monitoring of the target weight may be performed. This may be by the differential weight of the bowl or other means. If the first threshold (which may be dependent on ingredient type) is reached [3120], then the speed may be reduced to S2 [3130] and the weight continues to be monitored. If the target weight is reached, the Threshold based speed routine may conclude with End [3199]. More than two dispensing motor rotation speeds may be utilized depending on engineering choices and ingredient type and processing (sliced, chopped, etc.). It will be clear to one skilled in the part that speed S2 may be set to higher than S1 in some cases where higher speed may give slower, more controlled dispensing.

(97) A Threshold Based Weight Measurement Frequency algorithm and software program may be disposed in the food making machine apparatus, and may select and control dispense rates/speed as a function of dispensed food weight sampling (for example, increase weight sample rate more frequently near target weight, etc.). The sample rate of that weight may be adjusted in a certain manner, for example, inversely proportional to the percentage of the target weight, and so on), and may be different (speed control and weight sampling rate) for various ingredients (for example; iceberg lettuce, spinach, carrots, nuts, raisins, seeds, croutons, etc.) which may require different algorithms to control the appropriate machine sub-units and/or components. The algorithm may stop the dispense if the target weight is achieved. As illustrated in FIG. 32, an illustrative example of a Threshold Based Weight Measurement Frequency algorithm and software program is shown in an overview flowchart. For example, the Start [3200] may begin the algorithm and a default weight sampling setting W1 of the food product dispensed [3210] may be set to dispense the food, whilst monitoring of the target weight may be performed. This may be by the differential weight of the

bowl or other means. If the first threshold (which may be dependent on ingredient type) is reached [3220], then the weight sampling may be increased or otherwise made in a more accurate manner to W2 [3230] and the weight continues to be monitored. If the target weight is reached, the Threshold based weight measurement routine may conclude with End [3199]. and may conclude with End Otherwise the dispensing continues with the more accurate W2 weight sensing scheme.

(98) An Ingredient Level Based Dispensing algorithm and software program may be disposed in the food making machine apparatus, and may select and control dispense rates/speed as a function of dispensed food levels in the food ingredient containers (for example, may need to increase the flipper rotation speed to dispense same amount of the food ingredient in same time when the level in the ingredient container is 25%, etc.). The rate or adjustment may be for various ingredients (for example; iceberg lettuce, spinach, carrots, nuts, raisins, seeds, croutons, etc.) at various container levels (for example, such as 100%, 75%, 50%, 33% 25%, 10%, 5%) which may require different algorithms to control the appropriate machine sub-units and/or components. As illustrated in FIG. 33, an illustrative example of an Ingredient Level Based Dispensing algorithm and software program is shown in an overview flowchart. For example, the Start [3300] may begin the algorithm and a default dispensing algorithm 1 [3310] may be activated and the level of food in the specific canister/container is monitored (usually by weight dispensed and kept track of in the software; however, also could be monitored by sensors, such as optical or proximity). If a first threshold of the canister is depleted, for example 33% [3320] then a second dispensing algorithm may be utilized to maintain accurate and precise food product dispense, for example, Dispensing algorithm 2 [3330] If the canister is now depleted to another threshold, for example, 66% [3340], then a third algorithm may be controlling the dispense such as dispensing algorithm 3 [3350]. The Ingredient Level Based Dispensing algorithm may conclude with End [3399].

(99) A Liquid Pullback algorithm and software program may be disposed in the food making machine apparatus, and may select and control the dispense of liquids (for example, salad dressings, etc.). Liquid can sometimes drip from dispenser after it has been stopped. Reversing the flow in the liquid dispenser may reduce unwanted drips. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 34, an illustrative example of a Liquid Pullback algorithm and software program is shown in an overview flowchart. For example, the Start [3400] may begin the algorithm and a default liquid dispensing algorithm 1 [3410] may be activated to dispense the desired liquid and execute a default pullback, which may include, for example, a time increment or number of reverse rotations, etc., depending on the type of dispense machinery. If drips are detected [3420], for example by a weight increase in-between salads made or other means, such as visually reported by a customer, the liquid pullback may be increased [3430] for that specific liquid and dispense machinery combination. For example, the viscosity of salad dressing may change with batch, or as the dispense container is near the end of it's dispense volume (liquid ageing/evaporation), or temperature excursions, etc. The pullback change may be sent to the dispense algorithm so that adjustments may be made to maintain a consistent product delivery volume. The Liquid Pullback algorithm may conclude with End [3499].

(100) A Dispenser Collision Recovery algorithm and software program may be disposed in the food making machine apparatus, and may select and control the dispenser which may jam due to misalignments or out of spec food (for example, larger diameter nuts than expected, clumps of spinach, etc.). When a jam is detected, the algorithm may re-center the dispenser and switch movement algorithms. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 35, an illustrative example of a Dispenser Collision Recovery algorithm and software program is shown in an overview flowchart. For example, the Start [3500] may begin the algorithm and a default dispensing algorithm 1 [3510] may be activated to dispense the food ingredient. If a jamming is detected in the food dispense [3520], then a unjamming dispensing algorithm 2 [3530] may be activated to attempt to unjam the container and dispenser. For example, the dispensing algorithm 2 may re-center the canister/container and switch dispenser

movement algorithms. It will be clear that many other types of unjamming algorithms may be possible. The Dispenser Collision Recovery algorithm may conclude with End [3599].

(101) An Ingredient Jamming Recovery Reverse Direction algorithm and software program may be disposed in the food making machine apparatus, and may when the amount of an ingredient being dispensed is less than expected (or otherwise an improper dispense is detected), reverse direction of the paddle(s) to break up a blockage (for example, larger diameter nuts than expected, clumps of spinach, etc. Food can get stuck in a container, leaving a void where paddles cannot reach).

Reversing direction may also comprise rapid forward and reverse motions, rapid reverse and slow forward, and other combinations including time, rotational acceleration and velocity. This recovery algorithm may also be combined with the Ingredient Jamming Recovery Carousel Shake algorithm herein to clear an ingredient jam. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 36, an illustrative example of an Ingredient Jamming Recovery Reverse Direction algorithm and software program is shown in an overview flowchart. For example, the Start [3600] may begin the algorithm and a default dispensing algorithm 1 [3610] may be activated to dispense the food ingredient. If a jamming is detected in the food dispense [3620], then a unjamming dispensing algorithm 2 [3630] may be activated to attempt to unjam the container and dispenser. For example, the dispensing algorithm 2 may reverse the rotation direction, which may include various velocity and acceleration changes. The Ingredient Jamming Recovery Reverse Direction algorithm may conclude with End [3699].

(102) An Ingredient Jamming Recovery Carousel Shake algorithm and software program may be disposed in the food making machine apparatus, and may when the amount of an ingredient being dispensed is less than expected (or otherwise an improper dispense is detected), shake the carousel back and forth to break up a blockage (for example, larger diameter nuts than expected, clumps of spinach, etc. Food can get stuck in a container, leaving a void where paddles cannot reach).

Shaking may also comprise rapid forward and reverse motions, rapid reverse and slow forward, and other combinations including time, linear/rotational acceleration and velocity. This recovery algorithm may also be combined with the Ingredient Jamming Recovery Carousel Shake algorithm herein to clear an ingredient jam. A zeroing of the container position would be performed to avoid a future error. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 37, an illustrative example of an Ingredient Jamming Recovery Carousel Shake algorithm and software program is shown in an overview flowchart. For example, the Start [3700] may begin the algorithm and a default dispensing algorithm 1 [3710] may be activated to dispense the food ingredient. If a jamming is detected in the food dispense [3720], then a unjamming dispensing algorithm 2 [3730] may be activated to attempt to unjam the container and dispenser. For example, the dispensing algorithm 2 may shake the carousel, which may include various velocity and acceleration changes, for example, back and forth movements. The Ingredient Jamming Recovery Carousel Shake algorithm may conclude with End [3799].

(103) A Fallback Container algorithm and software program may be disposed in the food making machine apparatus, and may when an ingredient runs out a second container can be utilized by switching the container that is used for that ingredient to the fallback dispenser for that ingredient. A message may be sent to the appropriate person or device to notify them of an empty container. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 38, an illustrative example of a Fallback Container algorithm and software program is shown in an overview flowchart. For example, the Start [3800] may begin the algorithm and as a dispense is occurring from a canister [3810], the canister is detected to have run out of an ingredient [3820]. This detection may be by various means, for example, such as by calculation, weight measurement, sensors, etc. Then the algorithm may direct the apparatus to move to a fallback canister with the same ingredients, if available [3830]. If not available, then a signal is sent to the appropriate machine caretaker to promptly refill the specific containers. The Fallback Container algorithm may conclude with End [3899].

(104) A Rocking Motion Dispense algorithm and software program may be disposed in the food making machine apparatus, and may when the machine zeros it can inadvertently drop ingredients and may be directed to oscillate the dispenser back and forth to keep the back side clear as much as possible. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 39, an illustrative example of a Rocking Motion Dispense algorithm and software program is shown in an overview flowchart. For example, the Start [3900] may activate the algorithm during a zeroing of a canister/container [3910]. If a drop of ingredients is detected during the zeroing [3920], then the dispenser for that container may be oscillated back and forth [3930] to clear the backside of the dispenser. The Rocking Motion Dispense algorithm may conclude with End [3999].

(105) A Bidirectional Motion algorithm and software program may be disposed in the food making machine apparatus, and as some ingredients may tend to jam in the container/cylinder if only dispensed in one direction, the machine may be directed to rotate the dispenser in one direction for a number of cycles, then go back the other direction for a number of cycles. More than two directional changes may be employed to mitigate the jams. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 40, an illustrative example of the Bidirectional Motion algorithm in action is shown wherein the motion of the dispenser paddles may be rotated clockwise [4010] for a few dispenses and then counterclockwise [4020] for another few dispenses. The precise numbers of dispenses will depend upon engineering judgements and decision making as well as the specific type of food ingredient.

(106) A Switch Directions Between Salads algorithm and software program may be disposed in the food making machine apparatus, and as some ingredients may tend to jam in the container/cylinder if only dispensed in one direction, the machine may be directed to rotate the dispenser in the opposite direction each time the ingredient is chosen to be deposited. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 41, an illustrative example of a Switch Directions Between Salads algorithm and software program is shown wherein the motion of the dispenser paddles may be rotated clockwise [4110] for a making one or more salads and then counterclockwise [4120] for making the next salad, or more than one. The precise number of salads made between each rotation direction change will depend upon engineering judgements and decision making.

(107) A Unidirectional algorithm and software program may be disposed in the food making machine apparatus, and as the default dispenser motion, the machine may be directed to rotate the dispenser in a single direction until target weight is reached. This algorithm may control the appropriate machine sub-units and/or components. This algorithm may be the default dispense algorithm wherein the dispenser is rotated in one direction (clockwise or counterclockwise) until the target weight is achieved.

(108) A Quantized weight dispense algorithm and software program may be disposed in the food making machine apparatus, and when dispensing medium to large amounts of ingredients the total time to dispense can be made shorter by moving the dispenser by a large angle before checking the dispensed weight. The dispenser may be rotated by a specific distance (learned or pre-determined to each specific food ingredient) before checking for weight. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 42, an illustrative example of a Quantized weight dispense algorithm and software program is shown in an overview flowchart. For example, the Start [4200] may activate the algorithm and rotate the paddle by x degrees [4210]. If the target weight is achieved [4220] then the algorithm may conclude with End [4299]. If the target weight is not achieved [4220], then the paddle may be rotated by a new amount of rotation.

(109) A Multiple Ingredient Dispensing algorithm and software program may be disposed in the food making machine apparatus. The time to make a salad can be reduced by depositing 2 ingredients nearly simultaneously. The apparatus/machine has two concentric rings of ingredients,

so the machine may dispense 2 ingredients simultaneously. This may be accomplished by sending multiple ingredient commands at a time, stored in a buffer, and if ingredients exist on the inner and outer rings of the same segment, dispense them both at the same time. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 43, an illustrative example of a Multiple Ingredient Dispensing is shown. Outer canister **4310** and inner canister **4320** may be positioned wherein both may dispense into the product bowl underneath (not shown), thus allowing two food ingredients to be dispensed nearly simultaneously, thus saving product (salad) manufacturing time.

(110) A Mapping Ingredient Locations to minimize time algorithm and software program may be disposed in the food making machine apparatus. The time that it takes to switch ingredients increases the amount of time it takes to make a salad. The apparatus/machine may tell the loader to arrange the ingredients in an order that minimizes the time it takes to make an average salad by using historical data regarding ingredients used in some chosen time period. For example, the time period may be a day, a week, 3 weeks, 6 weeks, two months; and may also track by day of the week (such as optimum Monday and Friday canister/container arrangements may be different) or by a local calendar. This algorithm may control the appropriate machine sub-units and/or components.

(111) A Predictive Zero Mechanism Undershoot algorithm and software program may be disposed in the food making machine apparatus. When the machine zeros it can inadvertently drop ingredients. The apparatus/machine may determine the average amount of weight that is being deposited during zeroing step and reduce target weight accordingly. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 44, an illustrative example of a Predictive Zero Mechanism Undershoot algorithm and software program is shown in an overview flowchart. For example, the Start [**4400**] may activate the algorithm and a dispense from a canister [**4410**] may be performed. The algorithm may determine if enough weight of the food ingredient was dispensed such that when the canister is zeroed, then the target weight will be achieved [**4420**] (due to zeroing the dispenser may drop additional food material from the canister). If so, then the Predictive Zero Mechanism Undershoot algorithm may conclude with End [**4499**].

(112) A Predictive Dispensing Undershoot algorithm and software program may be disposed in the food making machine apparatus. The current method of feedback uses a scale to measure weight, once a weight measurement exceeds the target weight, the algorithms are stopped. This means that substantially all of the final weights will skew high. The apparatus/machine may determine the average amount of weight that is being deposited per rotation and stop if the weight will exceed target on the next rotation. This algorithm may control the appropriate machine sub-units and/or components. As illustrated in FIG. 45, an illustrative example of a Predictive Dispensing Undershoot algorithm and software program is shown in an overview flowchart. For example, the Start [**4500**] may activate the algorithm and a dispense from a canister [**4510**] may be performed. The algorithm may determine if enough weight of the food ingredient was dispensed such that when the dispense is stopped, then the target weight will be achieved [**4520**] (due to next rotation of the dispenser may drop additional food material from the canister). If so, then the Predictive Dispensing Undershoot algorithm may conclude with End [**4599**].

(113) An Ingredient Specific Undershoot algorithm and software program may be disposed in the food making machine apparatus. The current method of feedback uses a scale to measure weight, once a weight measurement exceeds the target weight, the algorithms are stopped. This means that substantially all of the final weights will skew high. Each ingredient seems to have a different amount of overshoot error to it. The machine/apparatus and program will quantify it and undershoot by that amount. The amount of typical undershoot for every ingredient will be measured there would be an Undershoot value in the gcode. Thus code G1 W50 U10 would stop at 40 g, since it would be expected that the overshoot would make up the difference. This algorithm may control the appropriate machine sub-units and/or components. This algorithm would work in a

similar fashion as FIG. 45 Predictive Dispensing Undershoot but be ingredient specific.

(114) A Predictive Dispensing Undershoot Using Aggregate Historical Data algorithm and software program may be disposed in the food making machine apparatus. Weight sensor measurements can take time and slow down dispensing. The machine/apparatus and program may use historical data to get near completion before dialing in the final target. This algorithm may control the appropriate machine sub-units and/or components.

(115) A Delayed Weight Measurement algorithm and software program may be disposed in the food making machine apparatus. The ingredients can be in the air when bowl is weighed. The machine/apparatus and program may wait an increment of time before measurements are taken. This will slow down the overall process. This algorithm may control the appropriate machine sub-units and/or components.

(116) An Automated Scale Calibration algorithm and software program may be disposed in the food making machine apparatus. Weight sensor measurements are important to delivery of an accurate to ordered salad. The scale may not be calibrated correctly and may not be giving accurate readings. The machine/apparatus and program may calibrate the weight sensor with known weights—for example, using the known bowl weight to calibrate. This algorithm may control the appropriate machine sub-units and/or components.

(117) FIG. 46A-D describe an embodiment of this invention, wherein a canister **4601** may be used to dispense a liquid, such as dressing, water, milk, smoothies or any other ingredient compatible with this mechanism. The liquid may be placed within bottle **4605** which in turn may be placed in a certain position using supports **4606**, as indicated in FIG. 46C. Tubing **4607** may be used to transport the liquid into a peristaltic pump apparatus **4602**. The peristaltic pump mechanism **4602** may be actuated with a motor using apparatus and methods described earlier in this patent application. This actuation may happen using pin **4604** (shown in FIG. 46B) and a shaft that enters the peristaltic pump mechanism. Tubing may enter the peristaltic pump mechanism and liquid may be pinched off using rollers **4608**, as indicated in FIG. 46D. Weight sensor readings may be taken after different dispensing motions and feedback may be provided to the dispensing motor. The dispensing motor may be shared among multiple liquid dispenser canisters, which may produce benefits to the food making apparatus' weight, size and/or cost. It will be clear to one skilled in the art that several variations of the proposed embodiments may be possible. Several peristaltic pump designs may be possible. Several liquid dispenser design variations may be possible.

(118) FIG. 47A-C describe an embodiment of this invention, wherein a canister **4701** may contain a tabbed paddle **4702**. FIG. 47B indicates a potential structure of the tabbed paddle **4702** that was shown in FIG. 47A. The paddle may contain a hard or rigid core or center **4703**. It may also contain flexible fins. The fins may contain a thicker portion **4705** as well as a thinner portion **4704** for optimized dispensing. The fins may also contain tabs such as **4706** which may add friction between the paddle and the canister **4701**. This may have the helpful benefit of preventing the paddle from moving due to gravity or other forces and could therefore keep the pin **4708** from getting misaligned. Multiple tabs **4707** may be placed on the same paddle to produce different amounts of friction between the paddle and the canister walls. Depending on the tab materials, canister materials and size of the tab, a certain maximum speed may be recommended for paddle rotation of any dispensing algorithm. It will be clear to one skilled in the art that several variations of the proposed embodiments may be possible. Weight readings may be taken during dispensing and the motor motion may be automatically controlled to control the dispensing. The same motor may be used to rotate paddles in different canisters. Position sensors may be used to locate position of the actuator arm, as indicated in the embodiment described in FIG. 20A-C.

(119) FIG. 48A-C describe an embodiment of this invention, wherein a canister **4801** may use shufflers **4802** for dispensing ingredients that may not work perfectly with a gravity fed mechanism. FIG. 48B indicates that shuffler end **4803** may come in contact with the paddle **4804** as the paddle **4804** rotates. This, in turn, may move the shuffler and push the ingredient in the

canister so it falls downwards. It will be clear to one skilled in the art that several embodiments may be possible for design of the shuffler. FIG. 48C indicates one embodiment for the shuffler, wherein structures 4805 allow the shuffler to be placed on the sides of the canister and shuffler end 4807 may have a coating material so that the paddle 4804 is not damaged by the shuffler hitting it. It will be clear to one skilled in the art that several apparatus and methods may exist to use the paddle rotation to create movement at higher positions in the canister and dispense ingredients that may not work perfectly with a gravity fed mechanism.

(120) FIG. 49A-D describe an embodiment of this invention, wherein a pin-shaft mechanism snaps into the paddle 4902 of a canister 4901. FIG. 49C may indicate the pin-shaft mechanism, which may consist of a pin 4903/4905 as well as a shaft 4904 which may have an end 4906. The end 4906 may snap into the structure 4907 in FIG. 49D which may be referred to as a retainer ring. By application of a push-in force, end 4906 may snap into the retainer ring 4907. By application of a pull-out force, end 4906 may be pulled out of the retainer ring 4907. FIG. 49B shows a view of the pin end inside the retainer ring (view 4908). It will be clear to one skilled in the art that several variations of these embodiments are possible. Different materials may be used for the shaft and retainer rings. Different shapes may be used for the shaft and retainer rings too.

(121) FIG. 50 illustrates an embodiment of this invention, wherein a rocking motion dispense algorithm may be used with weight feedback. After the start of the algorithm 5000, the paddle may be rotated by a certain angle “x” in one direction, then rotated back to the center, then rotated by the angle “x” in another direction, then rotated back to the center. A weight measurement may be made after that step. If the target weight is reached, the algorithm ends 5099. Else, the rocking motion may continue with the same angle “x” repeatedly till the target weight is reached.

Alternatively, the rocking motion may have increasing values of angle “x” till the target weight is reached. It will be clear to one skilled in the art that the embodiment described in FIG. 50 may be combined with embodiments described earlier in this patent application.

(122) References to “one embodiment,” “an embodiment,” “example embodiment,” “various embodiments,” etc., may indicate that the embodiment(s) of the invention so described may include a particular feature, structure, or characteristic, but not every embodiment necessarily the particular feature, structure, or characteristic.

(123) Further, repeated use of the phrase “in one embodiment,” or “in an illustrative embodiment,” do not necessarily refer to the same embodiment, although they may. The various embodiments described herein may be combined and/or features of the embodiments may be combined to form new embodiments.

(124) Unless specifically stated otherwise, as apparent from the following discussions, it is appreciated that throughout the specification discussions utilizing terms such as “processing,” “computing,” “calculating,” “determining,” or the like, refer to the action and/or processes of a computer or computing system, or similar electronic computing device, that manipulate and/or transform data represented as physical, such as electronic, quantities within the computing system's registers and/or memories into other data similarly represented as physical quantities within the computing system's memories, registers or other such information storage, transmission or display devices.

(125) In a similar manner, the term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. A “computing platform” may comprise one or more processors.

(126) Embodiments of the present invention may include apparatuses for performing the operations herein. An apparatus may be specially constructed for the desired purposes, or it may comprise a general purpose device selectively activated or reconfigured by a program stored in the device.

(127) Embodiments of the invention may be used for making several types of food—salads, bowls, breakfast bowls, acai bowls, fruit bowls, smoothies, cocktails, frozen yogurt and many other types

of food.

(128) It will also be appreciated by persons of ordinary skill in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and sub-combinations of the various features described hereinabove as well as modifications and variations which would occur to such skilled persons upon reading the foregoing description. Thus the invention is to only be limited by the claims.

Claims

1. An automated food making apparatus, comprising: a canister; a paddle disposed within the canister, with the paddle being configured to rotate; a shuffler disposed within the canister, with the shuffler having a first end that extends to an opposite second end, wherein the first end of the shuffler is placed on a top side of the canister and the second end of the shuffler is configured to directly contact the paddle as it rotates, wherein the canister is configured to store and dispense ingredients, and wherein the shuffler is configured to push the ingredients stored in the canister.
2. The automated food making apparatus of claim 1, wherein the shuffler is placed along an inner side of the canister.
3. The automated food making apparatus of claim 1, wherein the second end of the shuffler has a coating material.
4. The automated food making apparatus of claim 1, further comprising: a second shuffler.
5. The automated food making apparatus of claim 1, further comprising: a gravity fed mechanism configured to dispense the ingredients.
6. The automated food making apparatus of claim 5, wherein the shuffler is configured to assist the gravity fed mechanism, wherein the shuffler pushes the ingredients stored in the canister.
7. The automated food making apparatus of claim 1, wherein the ingredients stored in the canister is lettuce.
8. The automated food making apparatus of claim 1, wherein the shuffler is one of a plurality of shufflers.
9. The automated food making apparatus of claim 1, further comprising a second shuffler, wherein the shuffler and the second shuffler are located on opposing sides of the canister.
10. An automated food making apparatus, comprising: a carousel having a plurality of canisters, each canister storing an ingredient; a shuffler located within at least one canister of the plurality of canisters, with the shuffler having a first end that extends to an opposite second end; and a paddle located within the at least one canister, wherein the paddle is configured to rotate to dispense the ingredient from the at least one canister, wherein the first end of the shuffler is placed on a top side of the at least one canister and the second end of the shuffler is configured to directly contact the paddle as it rotates, and wherein the shuffler is configured to assist the dispensing by pushing the ingredients stored in the canister.
11. The automated food making apparatus of claim 10, wherein the shuffler is coupled to an inner side of the at least one canister.
12. The automated food making apparatus of claim 10, wherein the second end of the shuffler has a coating material.
13. The automated food making apparatus of claim 10, further comprising a second shuffler, wherein the shuffler and the second shuffler are located on opposing sides of the at least one canister.
14. A method for dispensing ingredients from an automated food making apparatus, the method comprising: providing a canister to store ingredients; providing a paddle and a shuffler within the canister, with the shuffler having a first end that extends to an opposite second end; rotating the paddle to dispense the ingredients from the canister; and directly contacting the second end of the

shuffler with the paddle as the paddle rotates to push the ingredients within the canister downward; and dispensing the ingredients from the canister.

15. The method of claim 14, wherein providing the shuffler comprises placing the first end of the shuffler on a top side of the canister.

16. The method of claim 14, further comprising providing a coating material on the shuffler.

17. The method of claim 14, further comprising providing a second shuffler within the canister.

18. The method of claim 14, further comprising: dispensing the ingredients with a gravity fed mechanism.

19. The method of claim 18, further comprising: assisting the gravity fed mechanism with the shuffler by pushing the ingredients with the shuffler.

20. The method of claim 14, wherein the shuffler is placed along an inner side of the canister.
